

Tutorato di Algoritmi e Laboratorio 4

Codifica di Huffman

$f(a) = 3$, $f(b) = 10$, $f(c) = 33$, $f(d) = 1$,
 $f(e) = 5$, $f(g) = 7$, $f(h) = 2$

caratteri : $n = \underline{7}$ lunghezza $m = 64$

char	f	codifica fissa	codifica di H.
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0	a	3	000
1	b	10	001
2	c	33	010
3	d	1	011
4	e	5	100
5	g	7	101
6	h	2	110

TOT

$> 6 = 4 + 2 = 110$ servono 3 bit

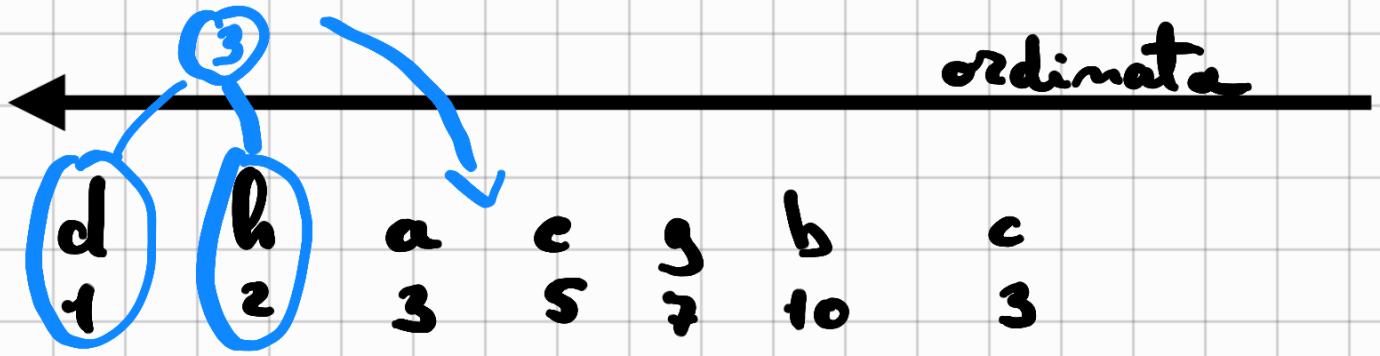
lunghezza $l = \lceil \log_2 n \rceil$

Ordinamento

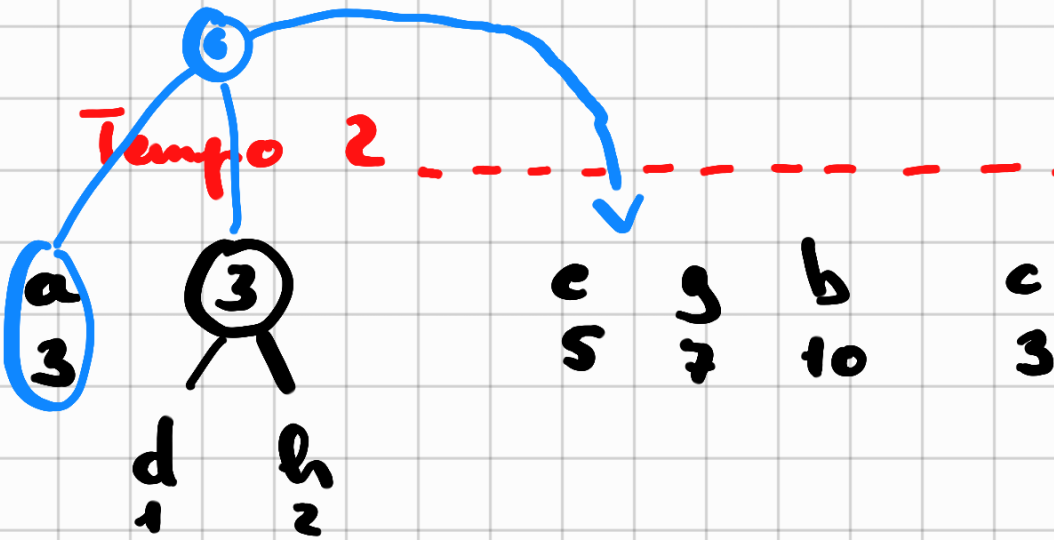
1) $1 < 2 < 3 < 4 < \dots < n-1 < n < n+1 < \dots$

2) $a < b < c < d < e < g < h$

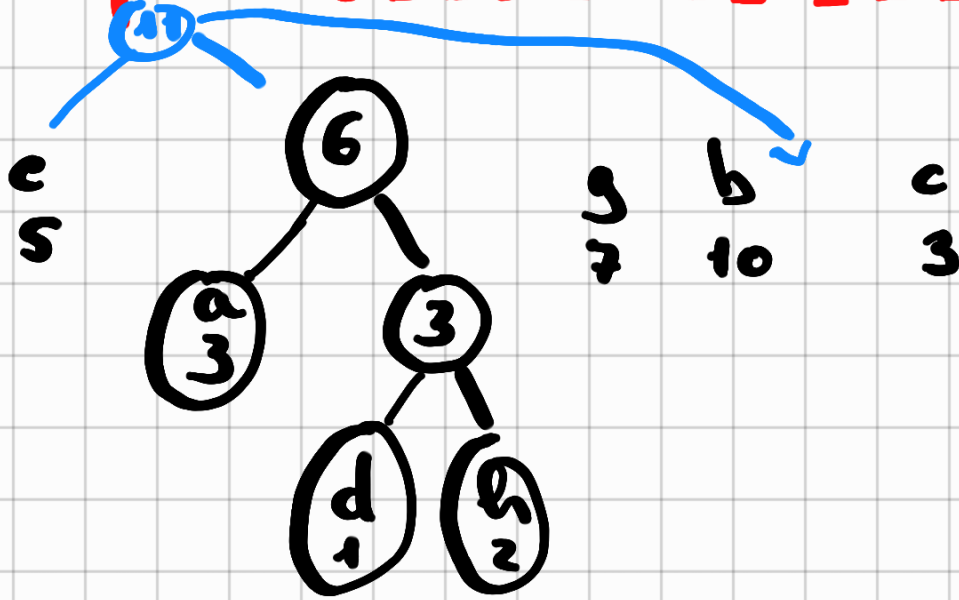
Tempo 1 - - - - -



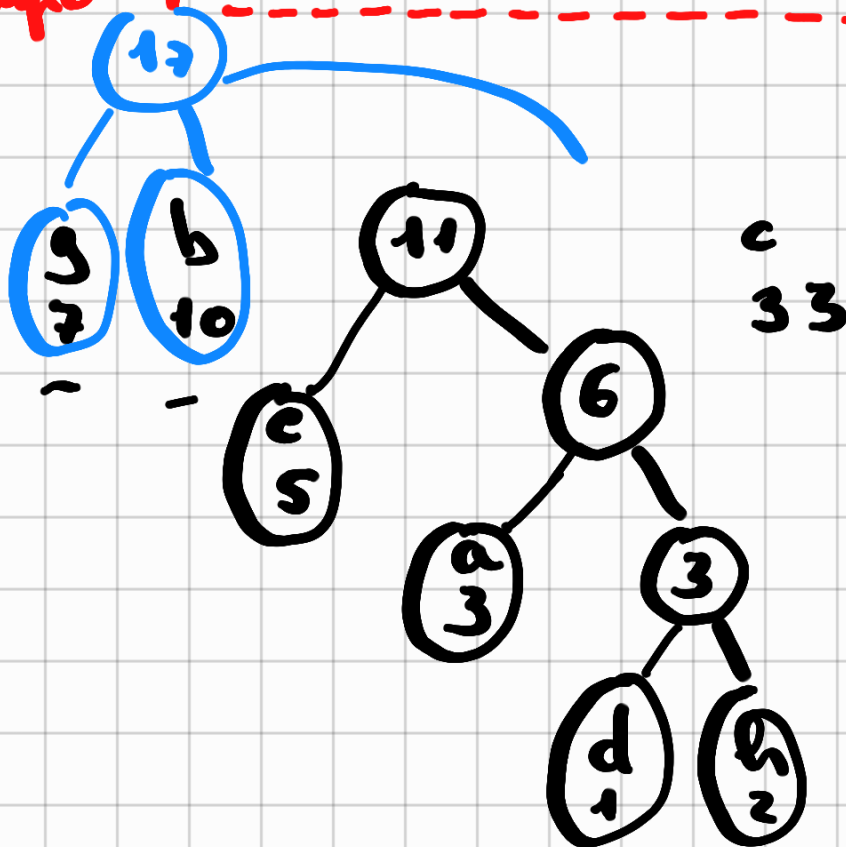
Tempo 2 - - - - -



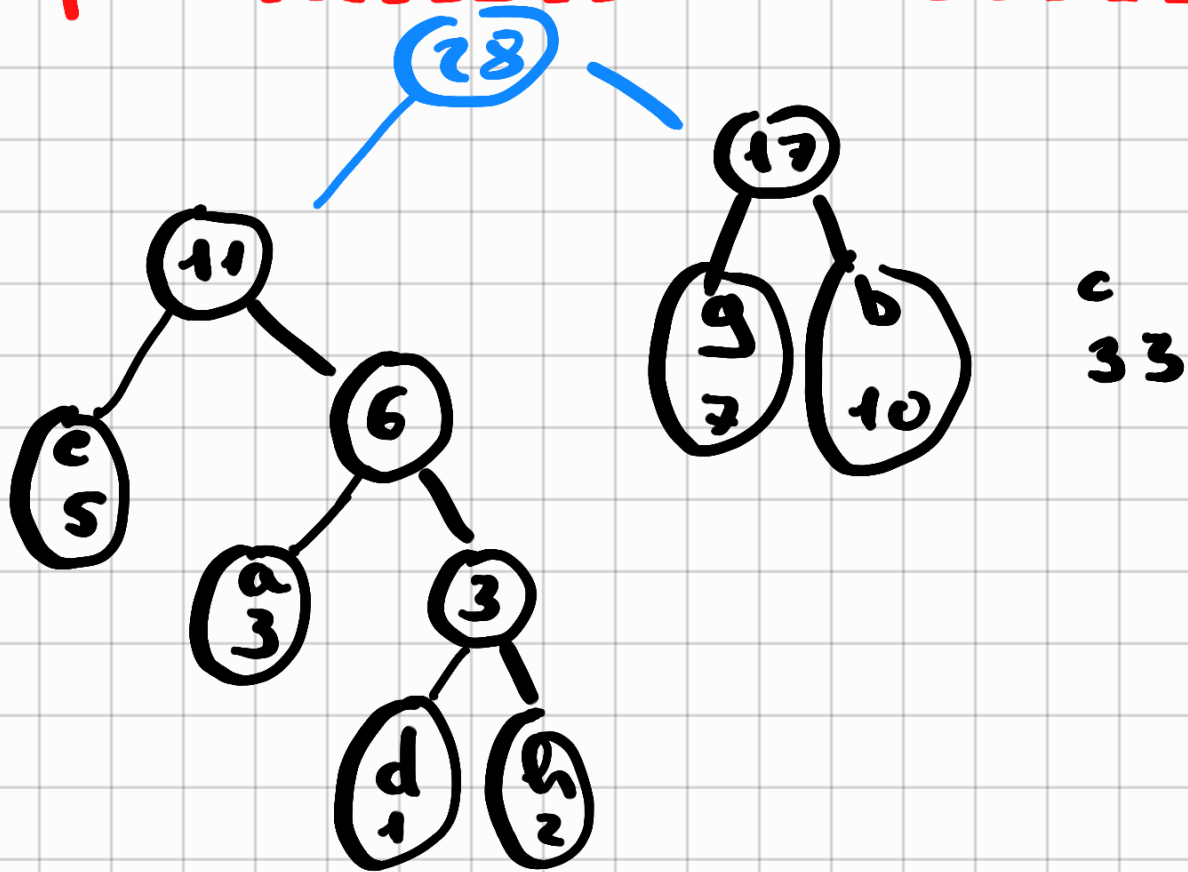
Tempo 3



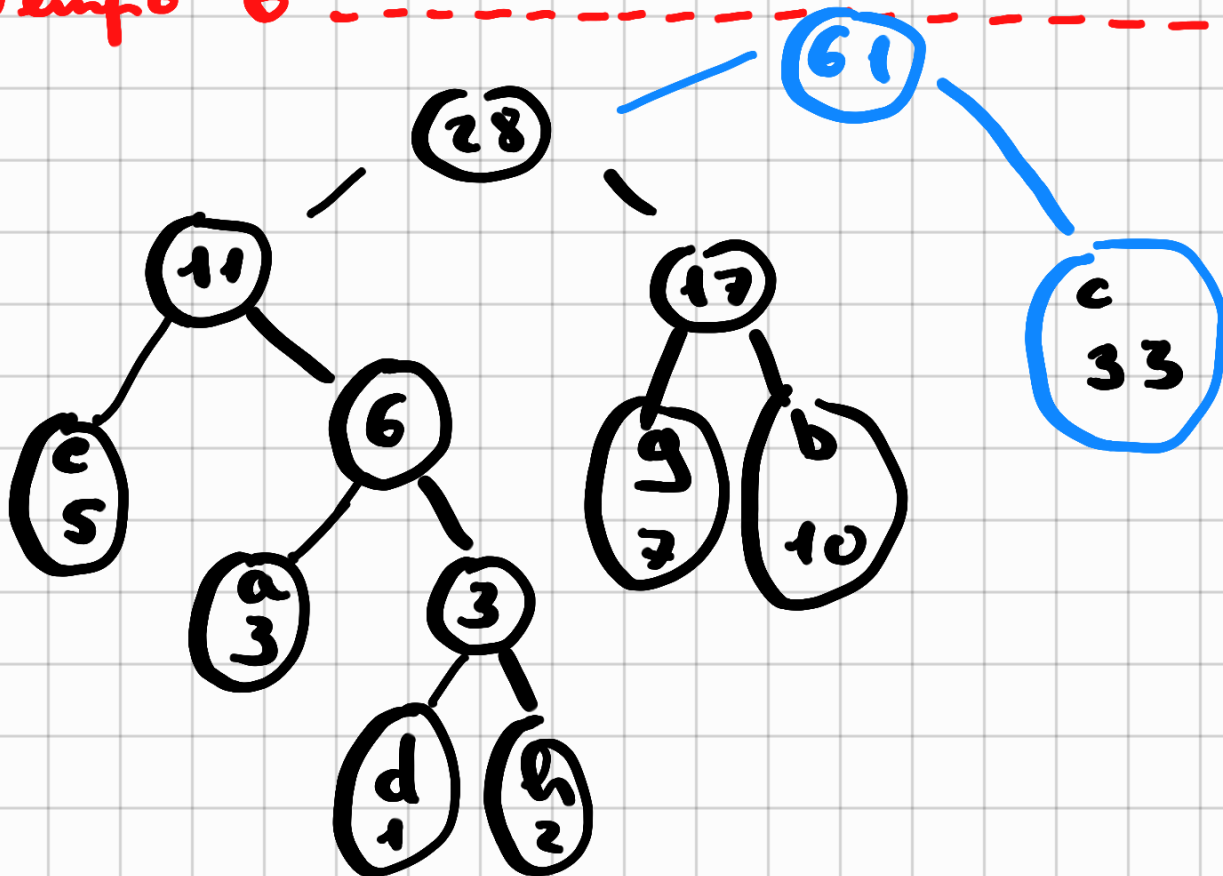
Tempo 4



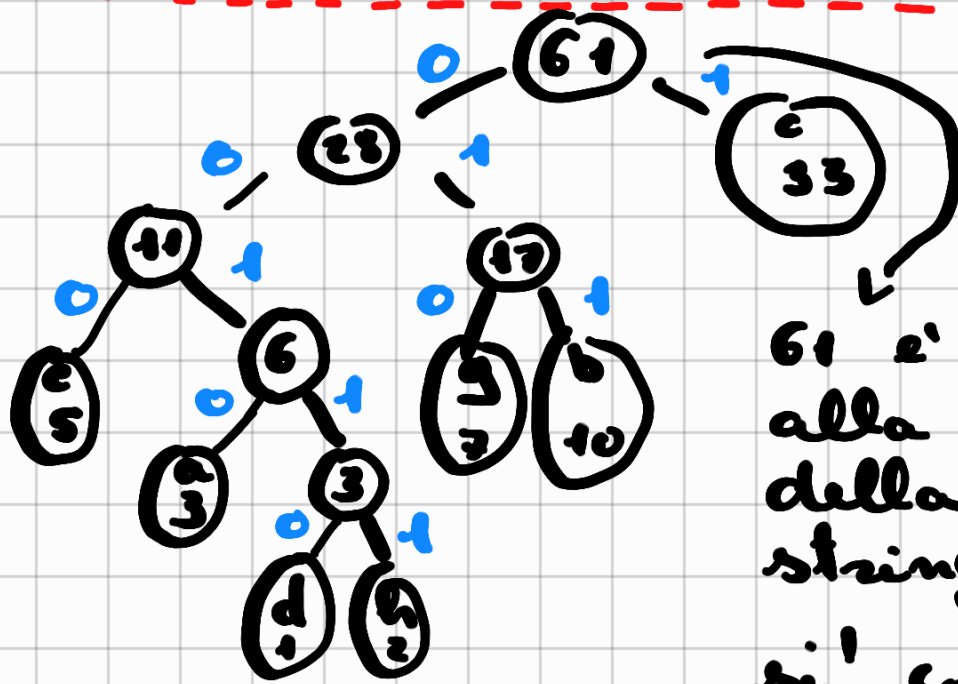
Tempo 5



Tempo 6



Tempo 7



61 e' uguale
alla lunghezza
della
stringa?
si! corretto!

bit per l'intero messaggio :

a lunghezza fissa : $m \cdot \lceil \log_2 n \rceil$

$$61 \cdot 3 = 183 \text{ bit}$$

con Huffman

= 126 bit

$$f(a) \cdot 4 + f(b) \cdot 3 + \dots$$

$$126 \text{ bit} \leq 183 \text{ bit}$$

vero!

Codifica di Huffman

a = 0010

b = 011

c = 1

d = 00110

e = 000

g = 010

h = 00111

Matrix Chain Multiply

$$p = \langle 2 \ 7 \ 2 \ 4 \ 3 \ 4 \rangle$$

$n = 6 - 1 = 5$ ovvero il numero di matrici

righe $= \langle 2 \ 7 \ 2 \ 4 \ 3 \rangle$ $\text{righe}_i = \# \text{ righe dell' } i\text{-esima matrice}$

colonne $= \langle 7 \ 2 \ 4 \ 3 \ 4 \rangle$ $\text{riga}_i = p_i$
 $\text{colonna}_i = p_{i+1}$

costo per moltiplicare la matrice i con la

matrice $i+1 = \text{righe}_i \cdot \text{colonne}_i \cdot \text{colonne}_{i+1}$

oppure $= \text{righe}_i \cdot \text{righe}_{i+1} \cdot \text{colonne}_{i+1}$

cosa è $A_{i,j}$ $=$ costo minimo della moltiplicazione della matrice i alla matrice j

costo per moltiplicare le matrici da i a j

mettendo la parentesi dopo K

$$= \underbrace{A_{i,K}}_{\text{colonna}_K} + \underbrace{A_{K+1,j}}_{\text{colonna}_{j+1}} + \text{righe}_i \cdot \text{colonna}_K \cdot \text{colonna}_j$$

oppure $=$

$$\text{righe}_i \left(\begin{pmatrix} i & K \end{pmatrix} \begin{pmatrix} K+1 & j \end{pmatrix} \right)$$

colonna_K colonna_{j+1} righe_{K+1}

$$\begin{array}{c}
 \text{red } 3 \\
 \text{blue } 2
 \end{array}
 \begin{pmatrix}
 \underline{1} & \underline{2} & \underline{1} \\
 1 & 1 & 3
 \end{pmatrix}
 \begin{array}{c}
 \text{red } 1 \\
 \text{blue } 3
 \end{array}
 \begin{pmatrix}
 \underline{1} \\
 \underline{2} \\
 \underline{3}
 \end{pmatrix}$$

$$\begin{array}{c}
 \text{red } 3 \\
 \text{blue } 1
 \end{array}
 \begin{pmatrix}
 1 \cdot 1 + 2 \cdot 2 + 1 \cdot 3 \\
 1 \cdot 1 + 1 \cdot 2 + 3 \cdot 3
 \end{pmatrix}$$

	0	1	2	3	4
0	0	28	44	64	88
1		0	56	66	104
2			0	24	48
3				0	48
4					0

$$p = \langle 2 \ 7 \ 2 \ 4 \ 3 \ 4 \rangle$$

$$A_{0,1} = A_{0,0} + A_{1,0} +$$

$$+ \text{right}_0 \cdot \text{column}_0 \cdot \text{column}_1 :$$

$$= 0 + 0 + 28 = 28$$

$$A_{1,2} = A_{1,1} + A_{2,1} + r_1 \cdot c_1 \cdot c_2 = 5$$

[...]

$$\begin{pmatrix} \overset{-}{i} & \overset{+}{j} \\ \begin{pmatrix} i \\ j \end{pmatrix} & \begin{pmatrix} 1 \\ 1 \end{pmatrix} \end{pmatrix}$$

$$A_{0,2} = \min \begin{cases} A_{0,0} + A_{1,2} + r_0 \cdot c_0 \cdot c_2 = 112 & \text{K=0} \\ A_{0,1} + A_{2,2} + r_0 \cdot c_1 \cdot c_2 = 44 & \text{K=1} \end{cases}$$

$$A_{1,3} = \min \begin{cases} A_{1,1} + A_{2,3} + r_1 \cdot c_1 \cdot c_3 = 66 & \text{K=1} \\ A_{1,2} + A_{3,3} + r_1 \cdot c_2 \cdot c_3 = 140 \end{cases}$$

$$A_{2,4} = \min \begin{cases} A_{2,2} + A_{3,4} + r_2 \cdot c_2 \cdot c_4 = 80 \\ A_{2,3} + A_{4,4} + r_2 \cdot c_3 \cdot c_4 = 48 & \text{K=3} \end{cases}$$

$$A_{0,3} = \min \begin{cases} A_{0,0} + A_{1,3} + r_0 \cdot c_0 \cdot c_3 = 108 \\ A_{0,1} + A_{2,3} + r_0 \cdot c_1 \cdot c_3 = 64 & \text{K=1} \\ A_{0,2} + A_{3,3} + r_0 \cdot c_2 \cdot c_3 = 68 \end{cases}$$

$$A_{1,4} = \min \begin{cases} A_{1,1} + A_{2,4} + r_1 \cdot c_1 \cdot c_4 = 104 & \text{K=1} \\ A_{1,2} + A_{3,4} + r_1 \cdot c_2 \cdot c_4 = 216 \\ A_{1,3} + A_{4,4} + r_1 \cdot c_3 \cdot c_4 = 150 \end{cases}$$

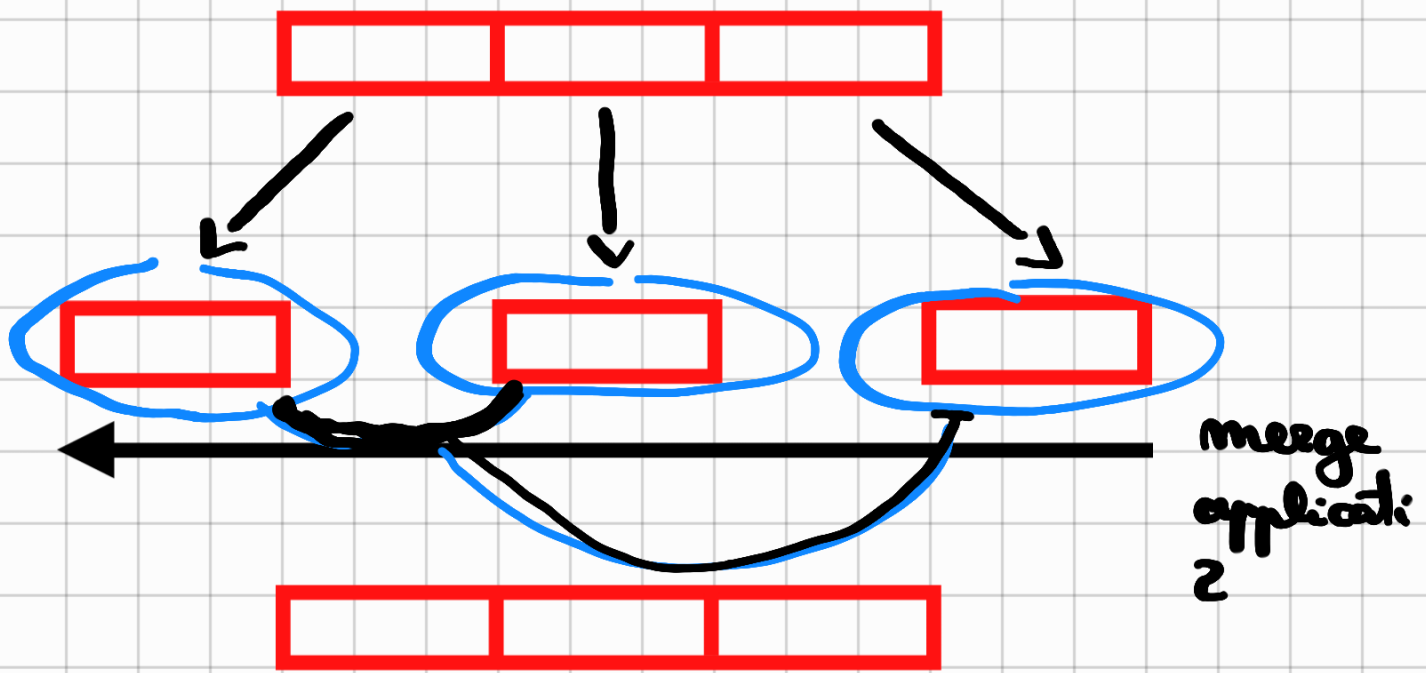
$$A_{0,4} = \min \begin{cases} A_{0,0} + A_{1,4} + r_0 \cdot c_0 \cdot c_4 = 160 \\ A_{0,1} + A_{2,4} + r_0 \cdot c_1 \cdot c_4 = 92 \\ A_{0,2} + A_{3,4} + r_0 \cdot c_2 \cdot c_4 = 174 \\ A_{0,3} + A_{4,4} + r_0 \cdot c_3 \cdot c_4 = 88 \end{cases}$$

$k=3$

$$\left(\left(A_0^0 \right) \left(A_1^1 \right) \left(A_2^2 \right) \left(A_3^3 \right) \left(A_4^4 \right) \right)$$

Ricorsione

Variante di Mergesort



$$T(n) \stackrel{?}{=} a T\left(\frac{n}{b}\right) + \Theta(g)$$

$$3 T\left(\frac{n}{3}\right) + \left(\left(\frac{n}{3} + \frac{n}{3}\right) + n\right)$$

$$3 T\left(\frac{n}{3}\right) + \Theta\left(n^1 \left(1 + \frac{2}{3}\right)\right)$$

$K=1$

$$\log_3 3 \stackrel{?}{=} 1$$

$$\log_3 3 = 1 \quad T(n) = \Theta(n \lg n)$$

Ricorsione

$$T(n) = a\left(\frac{n}{b}\right) + \Theta(\overbrace{n^3}^{n^3} + 2n + 1) \cdot \lg_b n$$

$$a = a$$

$$b = 4$$

$$K = 3$$

$$h = 0$$

$$\log_b a \stackrel{?}{\geq} K$$

$$\log_4 a \stackrel{?}{\geq} 3$$

↙

$$\log_4 a < 3$$

$$\log_4 a < 3 \cdot 1$$

$$\log_4 a < 3 \cdot \log_4 4$$

$$\log_4 a < \log_4 4^3$$

$$\log_4 a = 3$$

$$\log_4 a > 3$$

$$1 \leq a < 4^3 = 64$$

$$\Theta(n^3)$$

$$a = 64$$

$$\Theta(n^3 \log_2 n)$$

$$a > 64$$

$$\Theta(n^{\log_4 a})$$

$$O(n^5)$$

$$n^3 \stackrel{?}{\in} O(n^5)$$

$3 \leq 5$ vero! per $1 \leq a < 64$

$$n^3 \log_2 n \stackrel{?}{\in} O(n^5)$$

$3 \leq 5$ vero per $a = 64$

$$n^{\log_4 a} \stackrel{?}{\in} O(n^5)$$

$$\log_4 a \stackrel{?}{\leq} 5$$

$$\log_4 a \leq 5 \log_4 4$$

$$\log_4 a \leq \log_4 4^5$$

$$\text{vero per } \underline{4^4 < a \leq 4^5}$$

vero per $1 \leq a \leq 4^5$

Esercizi:

Codifica di Huffman

Dire quanti bit si risparmiano usando la codifica di Huffman e fornire una con i seguenti dati:

1) $f(a) = 1$ $f(b) = 1$ $f(c) = 7$
 $f(d) = 72$ $f(e) = 72$ $f(g) = 100$

2) $f(a) = 2$ $f(b) = 3$ $f(c) = 4$
 $f(d) = 5$ $f(e) = 4$ $f(g) = 3$
 $f(h) = 2$

Matrix Chain Multiply

Applicare M-C-M ai seguenti vettori e fornire la parentizzazione in base ai risultati attenti:

1) $\langle 10, 12, 13, 17, 19, 21 \rangle$

2) $\langle 4, 15, 5, 17, 3, 5 \rangle$

Ricorsioni

Analizzare

1) $T(n) = a T\left(\frac{n}{5}\right) + \Theta(n^5 + 2)$

2) $T(n) = 1 T\left(\frac{n}{6}\right) + \Theta(n^1 + 7)$

e dire quando esse risultano essere

- $O(n^4)$
- $\Theta(n^7)$
- $\Omega(n^2 \log_2 n)$
- $o(n^3 \log_2^2 n)$
- $\omega(n^6)$